

ChefVision AI

Project Workbook

By

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Chapter 1. Literature Search, State of the Art

Literature Search

ChefVision AI sits at the intersection of:

1. **Document understanding** (invoice/receipt extraction),
2. **Entity normalization** (ingredient name standardization / record linkage), and
3. **Forecasting + planning** (multi-horizon demand prediction + event-based ingredient planning).

Key research themes we are leveraging:

- **Visual Document Understanding (VDU) and Receipt Parsing**
 - Transformer-based document models reduce manual feature engineering and can learn layout + text jointly.
 - Receipt datasets and competitions provide benchmarks for line-item extraction and key field extraction.
- **Entity Matching / Normalization**
 - “Same entity, different name” is a core data management problem; deep learning + embeddings improve matching robustness over rules alone.
 - ChefVision’s normalization layer combines similarity scoring, rules (units/vendor patterns).
- **Forecasting (Short-Term + Multi-Horizon)**
 - Practical forecasting systems emphasize robustness, seasonality, and evaluation workflows.
 - Deep multi-horizon models (attention-based) can incorporate static covariates and known future inputs.
 - Gradient boosting remains strong for tabular forecasting features and cold-start regimes.

How the literature maps to our 3-model plan:

- **Model 1 (Transformer):** Document extraction from invoice/receipt images
- **Model 2 (ML):** Recipe–ingredient modeling for usage/costing knowledge (plus curated recipes)
- **Model 3 (ML + planning logic):** Forecasting + event-based ingredient requirements.

State-of-the-Art Summary

Modern “Document AI” has moved beyond basic OCR into **structured extraction** (tables, key-value pairs, line items) and **domain-specific parsers**. Leading-edge approaches increasingly combine:

- **Multimodal Transformers** (text + layout + vision) for robust extraction across formats
- **OCR-free or OCR-light pipelines** to reduce error propagation and simplify engineering
- **Human-in-the-loop review** for low-confidence fields to keep accuracy high at reasonable cost
- **Entity resolution + embeddings** to normalize noisy vendor naming and unify catalogs
- **Multi-horizon forecasting** to incorporate seasonality, local signals, and event schedules

In restaurant operations, existing inventory platforms primarily focus on inventory counts, ordering workflows, and recipe costing. ChefVision AI differentiates by making photo-first ingestion + ingredient normalization + event-based planning the center of the workflow:

- Start with photos/CSVs in minutes (low onboarding friction)
- Normalize supplier naming differences into a canonical ingredient catalog
- Convert events and recipes into ingredient requirement plans (week-by-week)
- Generate actionable reorder recommendations and shortage alerts

References

1. Geewook Kim, Teakgyu Hong, Moonbin Yim, JeongYeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, Seunghyun Park. **“OCR-free Document Understanding Transformer (Donut).”** *European Conference on Computer Vision (ECCV)*, 2022. (Also available as arXiv:2111.15664).
Description: A model that reads invoice/document images and extracts structured data without needing a separate OCR step. It converts raw images directly into structured JSON output.
2. Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, Furu Wei. **“LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking.”** arXiv:2204.08387, 2022. (arXiv)
Description: A pre-trained model that understands documents by learning from text, layout positions, and images together. Useful for extracting key fields from invoices where position on the page matters.
3. Sidharth Mudgal, Han Li, Theodoros Rekatsinas, AnHai Doan, Youngchoon Park, Ganesh Krishnan, Rohit Deep, Esteban Arcaute, Vijay Raghavendra. **“Deep Learning for Entity Matching: A Design Space Exploration.”** *SIGMOD 2018*, pp. 19–34. DOI: 10.1145/3183713.3196926. (DBLP)
Description: A study on using deep learning to decide whether two different text descriptions refer to the same real-world item. Relevant for matching supplier ingredient names to a standard catalog.
4. Bryan Lim, Sercan O. Arik, Nicolas Loeff, Tomas Pfister. **“Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting.”** arXiv:1912.09363, 2019. (arXiv)

Description: A forecasting model that predicts future values across multiple time steps using historical data and known future inputs. Can be used to forecast ingredient demand based on upcoming events and past usage patterns.

Chapter 2. Project Justification

ChefVision AI solves a common, high-impact operational problem: **inventory decisions are only as good as the data behind them**, and that data is often unstructured (paper invoices, inconsistent item naming, scattered notes).

Why this matters:

- **Financial impact:** Mis-ordering drives waste, margin loss, and staff time spent reconciling invoices manually.
- **Operational reliability:** Stockouts create lost sales and degraded customer experience, especially during demand spikes.
- **Complexity increases over time:** More suppliers, changing pricing, and varying naming conventions create long-term data fragmentation.

Why ChefVision AI is the right approach:

- **Photo-first ingestion** matches the real workflow (owners already receive paper invoices and take photos).
- **Normalization is the multiplier:** Once “tomato / tomatoh / roma tomato” becomes one canonical ingredient, every downstream function improves (inventory accuracy, costing, forecasting).
- **Event-based planning is actionable:** Restaurants often plan around catering/events; converting events + recipes into ingredient requirements creates immediate value even before forecasting matures.
- **Scales from MVP to product:** Start with ingestion + normalization + Excel outputs; expand to integrations and predictive analytics as data accumulates.

Success criteria:

- Reduce manual invoice processing time
- Improve ordering accuracy (fewer stockouts and less overstock)
- Provide clear, explainable reorder suggestions tied to inventory + planned events

Chapter 3. Project Requirements

Requirements (Use Cases / User Stories)

Essential (MVP)

- **Invoice Capture & Upload**
 - As a restaurant owner, I can take photos of invoices/receipts and upload them to the app.
 - As for the system, I can store original images and extracted results with timestamps and vendor metadata.
- **Document Extraction (Model 1)**
 - As for the system, I can extract vendor name, date, totals, and line items (ingredient name, qty, unit, unit price, total price).
 - As the restaurant owner, I can review extracted line items and correct errors.
- **Ingredient Normalization (Core AI)**
 - As for the system, I can detect likely duplicates and spelling variants (e.g., tomato vs tomotoh) and map them to one canonical ingredient.
 - As the restaurant owner, I can confirm or override suggested matches when confidence is low.
 - As the admin, I can manage the canonical ingredient catalog and synonym rules.
- **Inventory Sheet Output**
 - As the restaurant owner, I can export/refresh an Excel-compatible inventory ledger (or spreadsheet view) with cleaned items and quantities.
- **Event + Recipe Planning**
 - As the restaurant owner, I can create upcoming events and enter: event name/date, recipes/menu items, and serving count (e.g., 100 people).
 - As the system, I can compute total ingredient requirements per event and aggregate week-by-week.
- **Smart Reorder List**
 - As the system, I can compare current inventory vs event requirements and generate a reorder list for shortages.
 - As the restaurant owner, I can see reorder quantities and the ingredient breakdown driving each recommendation.
- **Roles / Two Faces**
 - As the platform admin, I can access admin-only configuration screens.
 - As a restaurant owner/user, I can access only my restaurant workspace, invoices, and planning tools.

Desired (Next Iteration)

- **Forecasting (Model 3)**
 - Predict short-term demand using historical usage + calendar effects; recommend purchase quantities for a defined horizon (e.g., 7–14 days).
- **Supplier/Vendor Analytics**
 - Track price history, detect price spikes, and suggest cheaper vendor alternatives (where data supports it).
- **Menu/Recipe Costing**
 - Compute per-recipe ingredient cost and trend it over time.
- **Integrations (Planned)**
 - POS and supplier integrations via API/file drops.

Non-Functional Requirements

- **Accuracy targets (initial):**
 - Line item extraction: target high precision on totals + quantities; allow user review for edge cases.
 - Normalization: high-confidence auto-merge; low-confidence requires confirmation.
- **Performance:**
 - Invoice processing completes within acceptable user wait time (or async with notifications).
- **Security & privacy:**
 - Role-based access control, encrypted storage, audit logs.
- **Reliability:**
 - Data integrity for inventory ledger; versioning for corrected extractions.
- **Usability:**
 - Photo upload should work with typical phone camera quality and imperfect lighting.

Chapter 4. Dependencies and Deliverables

Dependencies

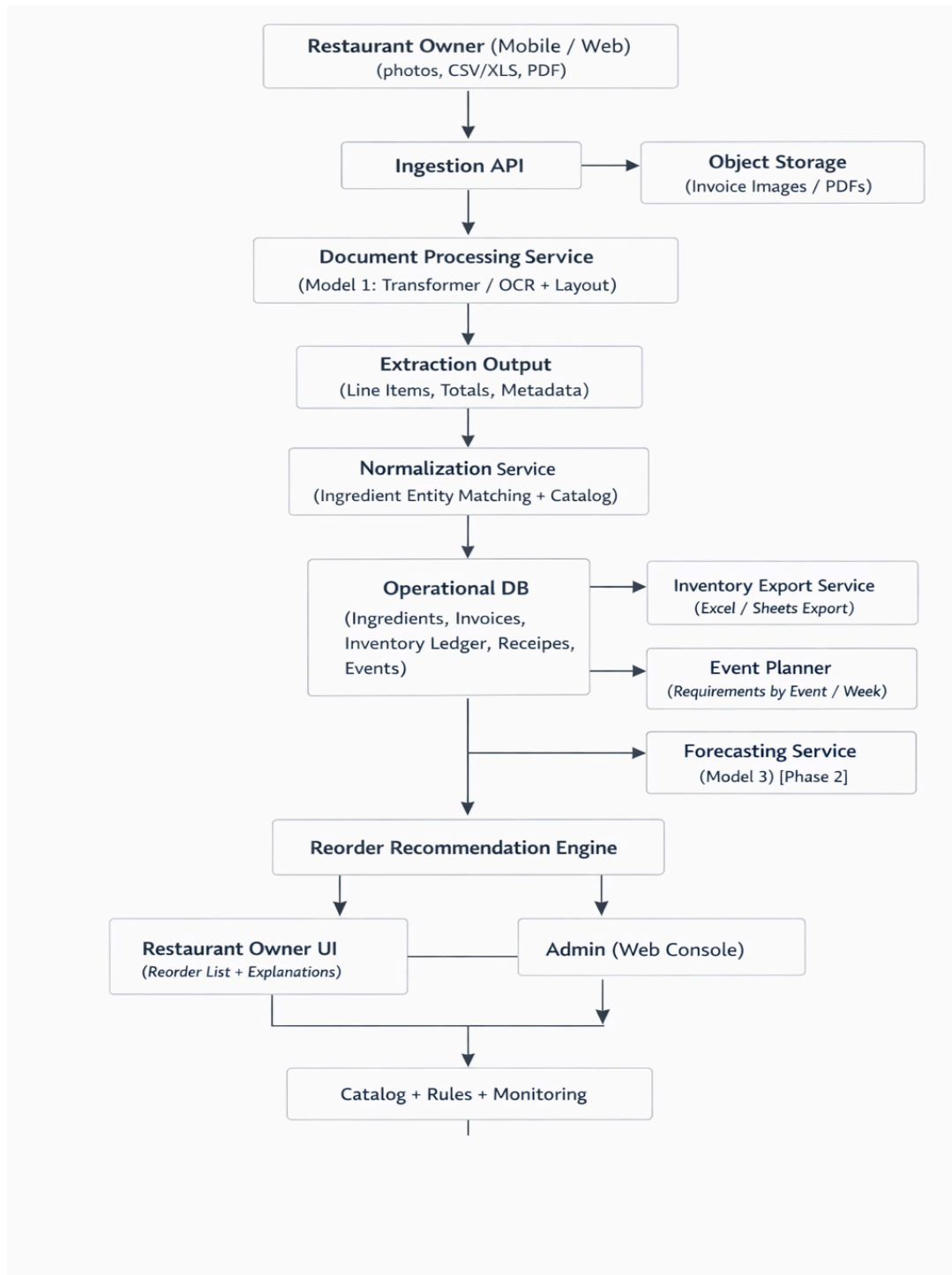
- Access to **representative invoices/receipts** for development/testing (real or sanitized).
- A **baseline ingredient catalog** (seed data) to support normalization early.
- Compute environment for model training/inference (GPU helpful for doc models).
- Storage for document images and extracted structured data.
- Recipe–ingredient dataset availability (public dataset + optional curated restaurant recipes).

Deliverables

- **Restaurant Owner App (User UI):** upload invoices, review extraction, inventory view/export, event planner, reorder list
- **Admin Console (Admin UI):** canonical ingredient catalog management, synonym rules, model/config controls
- **Backend API + Database:** authentication, data storage, business logic
- **ML Services**
 - Model 1: document extraction service
 - Normalization service: matching + confidence scoring
 - Model 3: forecasting (when enabled)
- Excel/Spreadsheet Export Template for inventory and planning outputs
- Demo Dataset + End-to-End Demo showing photo → inventory → event plan → reorder list.

Chapter 5. Project Architecture

Diagram



Description

ChefVision AI uses a client-server architecture. Users upload invoices as photos (primary ingestion) and optionally upload files (CSV/Excel/PDF). Documents are stored in object storage, then processed by a Document AI service (Model 1) that extracts structured line items. Extracted items are passed into a normalization service that resolves naming variants to a canonical ingredient catalog using embeddings + rules + confidence scoring. Cleaned inventory updates are stored in the operational database and can be exported to an Excel-compatible sheet.

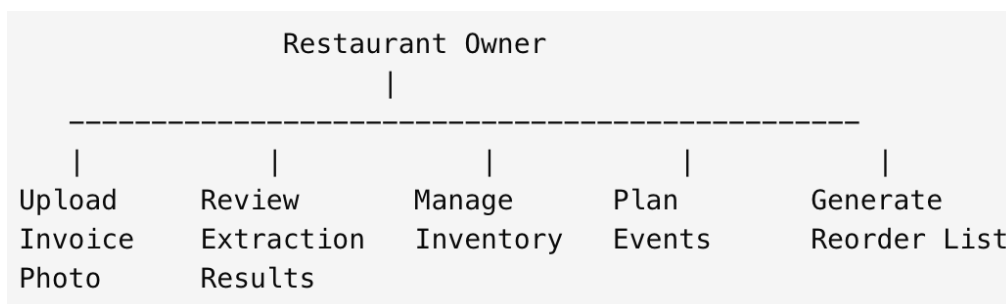
On top of this cleaned data layer, the application supports event planning: users specify events, recipes, and serving counts; the system calculates ingredient requirements and checks inventory sufficiency. If shortages exist, the reorder engine generates a smart reorder list with explainable quantities (tied to event plans and historical usage). In later phases, a forecasting service (Model 3) provides short-term demand estimates to refine reorder quantities beyond events.

Chapter 6. Project Design

Use Case Diagram

Actors:

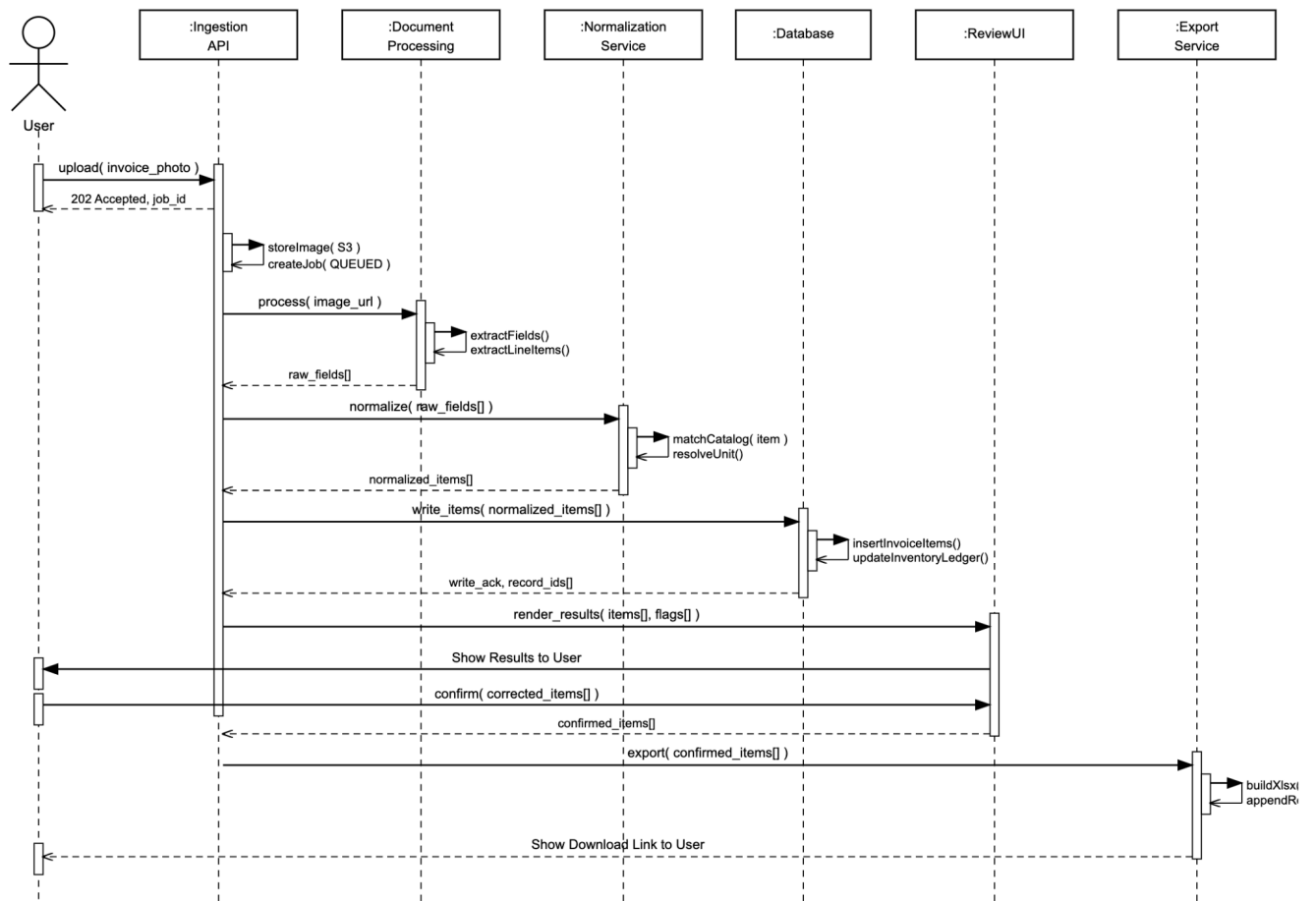
- Restaurant Owner



Use cases:

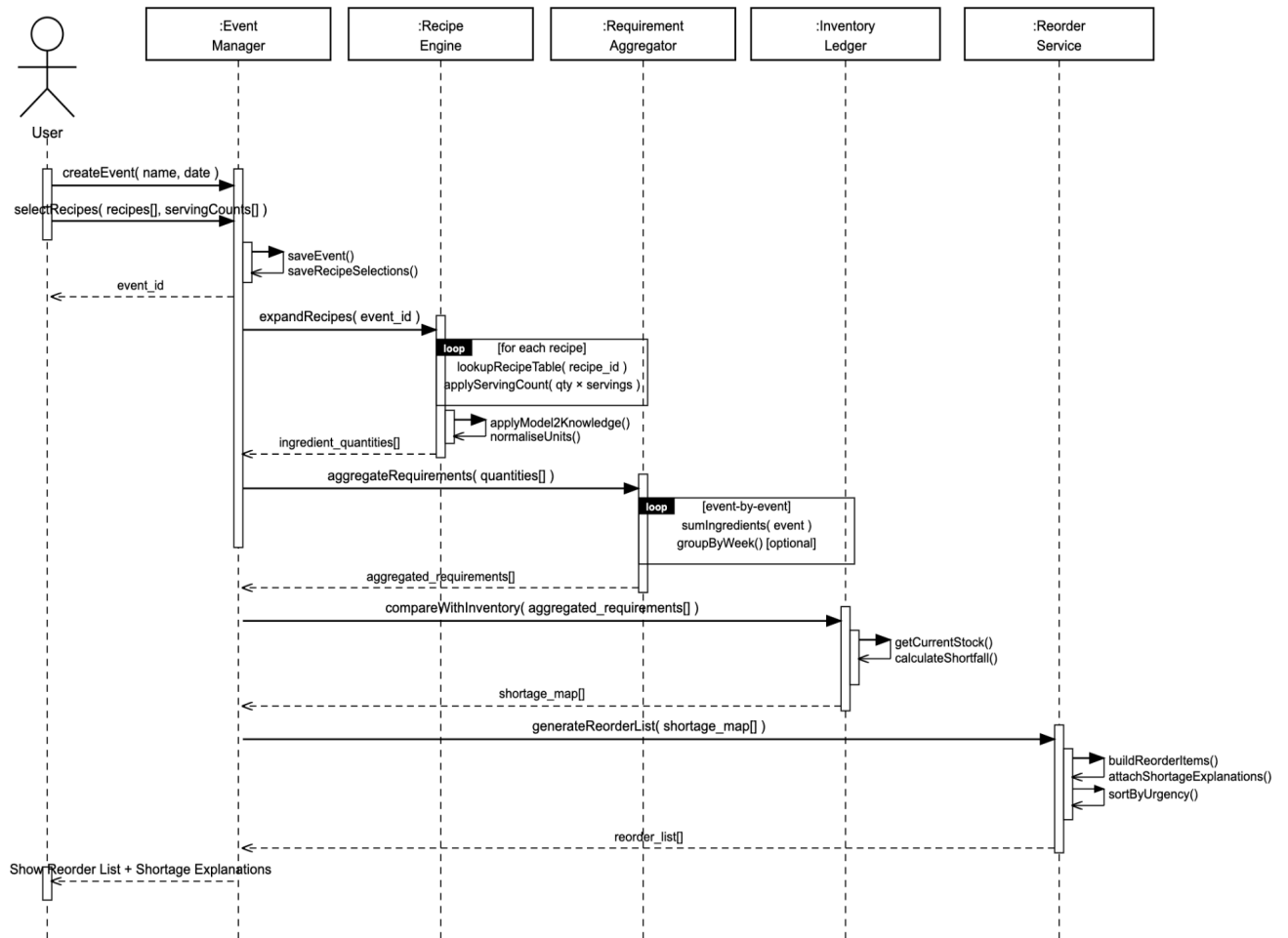
- UC1: Upload invoice photo
- UC2: Extract invoice line items (auto)
- UC3: Review/correct extracted items (human-in-loop)
- UC4: Normalize ingredient names to canonical catalog
- UC5: Export/refresh inventory spreadsheet
- UC6: Create event plan (recipes + servings)
- UC7: Compute ingredient requirements and shortages
- UC8: Generate reorder list and weekly planning summary
- UC9: Admin manages catalog + synonym rules + confidence thresholds

Sequence (Invoice → Inventory)



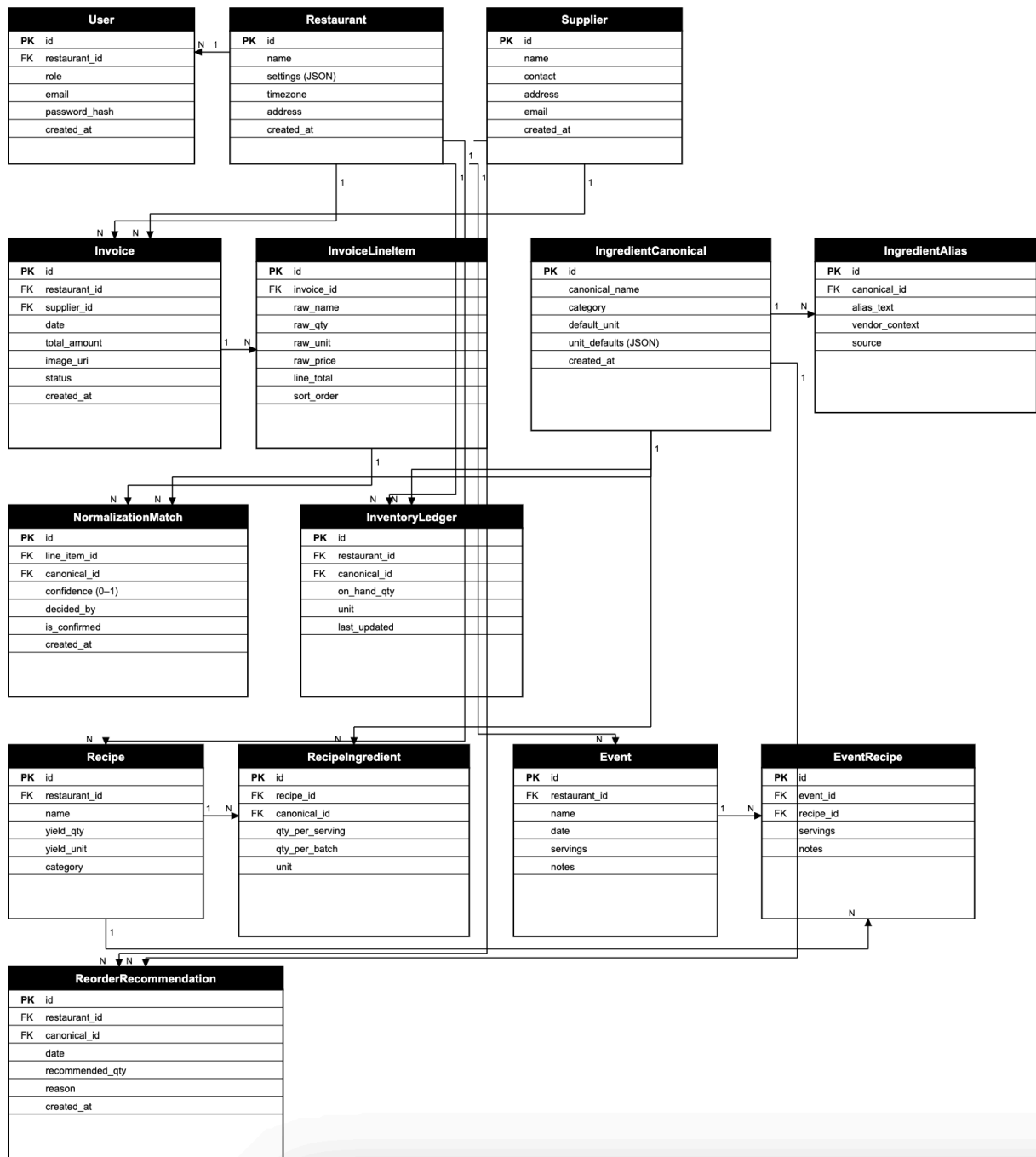
1. User uploads invoice photo
2. Ingestion API stores image, creates processing job
3. Document Processing extracts fields + line items
4. Normalization matches ingredients to canonical catalog
5. System writes cleaned items to DB and updates inventory ledger
6. UI shows results + lets user confirm corrections
7. Export service updates Excel output

Sequence (Event → Reorder List)



1. User creates events and selects recipes + serving counts
2. System expands recipes into ingredient quantities (Model 2 knowledge + recipe table)
3. System aggregates requirements event-by-event and optionally week-by-week
4. System compares requirements vs inventory ledger
5. System generates reorder list + shortage explanations

Data Model (Entities)



- User(id, role, restaurant_id, ...)
- Restaurant(id, name, settings, ...)
- Supplier(id, name, contact, ...)
- Invoice(id, restaurant_id, supplier_id, date, totals, image_uri, status, ...)
- InvoiceLineItem(id, invoice_id, raw_name, raw_qty, raw_unit, raw_price, ...)
- IngredientCanonical(id, canonical_name, category, unit_defaults, ...)
- IngredientAlias(id, canonical_id, alias_text, vendor_context, ...)

- NormalizationMatch(id, line_item_id, canonical_id, confidence, decided_by, ...)
- InventoryLedger(id, restaurant_id, canonical_id, on_hand_qty, last_updated, ...)
- Recipe(id, name, yield, unit, ...)
- RecipeIngredient(id, recipe_id, canonical_id, qty_per_serving or qty_per_batch, ...)
- Event(id, restaurant_id, name, date, servings, notes, ...)
- EventRecipe(id, event_id, recipe_id, servings, ...)
- ReorderRecommendation(id, restaurant_id, date, canonical_id, recommended_qty, reason, ...)

Chapter 7. QA, Performance, Deployment Plan

Testing Plan

- **Unit tests**
 - Normalization logic (string + embedding match, confidence thresholds)
 - Inventory math (units, conversions, aggregation)
 - Event requirement calculations and reorder list generation
- **Integration tests**
 - Photo upload → extraction job → normalized output in DB
 - Export generation correctness (Excel output matches DB state)
- **Model evaluation**
 - Document extraction: field-level precision/recall/F1 on a labeled invoice set
 - Normalization: match accuracy + human override rate
 - Forecasting (phase 2): MAPE/SMAPE over rolling backtests
- **User acceptance tests**
 - End-to-end workflow: invoice → cleaned inventory → event → reorder list

Performance Plan

- Measure processing time per invoice and identify bottlenecks (upload, extraction, normalization).
- Load testing for peak upload times (batch invoice processing).
- Ensure async processing works reliably with job status + retries.

Deployment Plan

- Containerized services (API, workers, ML inference) deployed to a cloud environment.
- CI/CD for automated test runs and staged deployments.
- Monitoring:
 - Logs, latency metrics, error rates
 - Model drift checks and extraction failure tracking
- Data backups and retention policies for invoices and ledger history.

Chapter 8. Implementation Plan and Progress

Implementation Plan

Phase 0: Setup

- Repo + branching strategy, environment setup
- DB schema draft, API skeleton, basic auth/roles

Phase 1: Ingestion + Extraction MVP

- Mobile/web upload flow
- Baseline extraction pipeline (Model 1 integration)
- Storage + job queue + retry logic
- UI for reviewing extracted line items

Phase 2: Normalization + Inventory Ledger

- Canonical ingredient catalog + alias table
- Matching logic
- Inventory ledger updates + Excel export

Phase 3: Recipe + Event Planning

- Recipe/ingredient tables + serving scaling
- Event planner UI
- Requirements computation and reorder list output

Phase 4: Forecasting

- Feature engineering from ledger + calendar + events
- Baseline models (boosting) then upgrade as data grows
- Backtesting + monitoring

Current Progress (as of March 4, 2026)

- Project scope defined: photo-first invoice ingestion → normalization → inventory sheet → event planning → reorder list.
- Three-model roadmap established (Document AI, recipe/ingredient model, forecasting + planning engine).

Chapter 9. Project Schedule

Sprint Plan

Sprint 0 (Mar 4 – Mar 10): Foundations

- Define MVP scope, user stories, acceptance criteria
- Set up repo, CI, dev environments, initial DB schema

Sprint 1 (Mar 11 – Mar 24): Ingestion + Job Pipeline

- Photo upload + storage
- Processing job queue + status tracking
- Stub UI for invoice list + upload

Sprint 2 (Mar 25 – Apr 7): Document Extraction (Model 1)

- Integrate extraction model/service
- Store extracted line items in DB
- Review UI (edit line items)

Sprint 3 (Apr 8 – Apr 21): Normalization + Catalog

- Canonical ingredient catalog + alias management
- Matching + confidence scoring
- “Confirm match” UI and audit trail

Sprint 4 (Apr 22 – May 5): Inventory Ledger + Excel Export

- Ledger updates from normalized line items
- Excel export generator + downloadable report
- Basic analytics (top spend, frequent ingredients)

Sprint 5 (May 6 – May 19): Event Planner + Reorder List

- Events + recipes + serving count
- Ingredient requirement calculator
- Reorder list generation (event-by-event + weekly summary)

Sprint 6 (May 20 – Jun 2): Hardening + Demo

- QA pass, performance tuning, bug fixes
- Final demo script and documentation

Team Contributions

- **Richard Pham:** Document extraction (Model 1), dataset prep, evaluation metrics
- **Girith Choudhary:** Ingredient normalization service, canonical catalog design, matching evaluation
- **Shivansh Chhabra:** Full-stack integration (UI/API), deployment pipeline, Excel export, testing harness